MASTER PRODUCT, ENGINEERING & LEGAL PLAYBOOK • EDITION 1.0

SSM Events & Ticketing Platform

A Complete Zero-Code Blueprint for Building a Hyperlocal Event Listing & Box-Office Ticketing Business in Sasaram (Rohtas) & Bihar • Written in Plain English for a Class 11 Commerce Founder

1. WELCOME TO YOUR FOUNDER'S PLAYBOOK (HOW TO READ THIS GUIDE)

This playbook is your step-by-step master manual to build, launch, and run **SSM Events & Ticketing**—a **Hyperlocal Platform** (a digital service built specifically for one local city or district rather than the whole country) serving **Sasaram, Dehri-on-Sone, Rohtas, Gaya, and Patna**. Because you are a **17-year-old Class 11 Commerce student**, you already have a superpower: you understand money, **Commission** (a percentage fee charged for helping make a sale), **Turnover** (total sales volume), and business trust. Whenever this playbook uses a technical or legal keyword, it is explained immediately inside parentheses *(just like this)* so you never feel confused.

2. THE PROBLEM: WHY BIG TICKETING APPS FAIL IN SASARAM & ROHTAS

National ticketing giants like **BookMyShow**, **Paytm Insider**, and **Zomato District** are designed for huge concerts in metro cities like Mumbai, Delhi, or Bengaluru. When local event organizers in Sasaram or Dehri-on-Sone try to sell tickets, they face four massive roadblocks:

- **High Commission Rates (10% to 15%):** National platforms take a huge cut of every ticket sold, which destroys the profit budget of a local college fest or coaching seminar.
- **Slow Corporate Onboarding:** Big apps ask small organizers for complex company papers, taking weeks to approve a single event listing.
- **Painful Login Friction for Buyers:** Local students and families in Sasaram hate downloading a heavy 80MB app, creating an account, typing an email password, and waiting for OTPs just to buy a ₹100 or ₹500 pass.
- **Delayed Payouts & No Cash Option:** Big platforms hold the organizer's money for 7 to 10 days after the event ends and do not support **Hybrid Sales** (selling tickets both online via UPI and offline via physical cash counters at the same time).

3. THE SOLUTION & WHY WE ARE BUILDING "HYPERLOCAL"

We are building a fast, mobile-first, **No-Login UPI Ticketing Platform** exclusively for Sasaram and Bihar. By focusing strictly on our own district first, we build a **Local Moat** (an unfair local business advantage that giant outside companies cannot easily copy because you personally know the coaching teachers, hotel owners, and college leaders in Sasaram).

1. Coaching Seminars & Scholarship Tests

Sasaram and Dehri are major education hubs. Coaching institutes regularly host career seminars, motivation talks, and ₹50–₹200 scholarship tests for 300 to 1,000 students.

2. New Year, Dandiya & Cultural Nights

Local hotels, marriage halls, and clubs organize high-ticket (₹500–₹1,000) Garba/Dandiya nights during Navratri, Holi milan events, and New Year parties.

3. School & College Annual Functions

Colleges and private schools in Rohtas host farewells, freshers' parties, tech fests, and cultural shows that need fraud-proof entry passes and student roll-number tracking.

4. Local Concerts, Standup & Workshops

Regional singers, Bhojpuri/Hindi poets, youth clubs, and skill academies hosting weekend workshops who want instant WhatsApp ticket sharing and fast bank payouts.

OUR CORE WINNING PROMISE TO SASARAM ORGANIZERS

Flat 7% Commission (vs. 15% elsewhere) • **Zero-Login 2-Click UPI Checkout** • **Instant QR Code Pass on Screen & WhatsApp** • **Fast T+2 Bank Settlements** *(money reaches bank 2 business days after payment)* • **₹0/Month Server Cost**.

4. UNIT ECONOMICS: THE 7% COMMISSION MATH (COMMERCE BREAKDOWN)

In Commerce, **Unit Economics** (the exact revenue, cost, and profit calculated on a single unit—in our case, one single ticket sold) decides whether a business survives. Your target is to charge a flat **7% Platform Commission** on every ticket sold, using **Razorpay** as our **Payment Gateway** (the regulated banking software that processes UPI, Debit Card, and NetBanking payments online).

Here is a crucial real-world tax detail every Commerce student must know: while Razorpay's standard fee is **2.00%**, the Indian government charges **18% GST** *(Goods and Services Tax)* on Razorpay's 2% service charge. 18% of 2% equals **0.36%**. So Razorpay automatically deducts **2.36% total**, leaving you with a clean **4.64% Net Profit Margin** on every ticket!

> **EXACT MONEY FLOW ON ONE ₹800.00 EVENT TICKET**
> **Base Ticket Price:** ₹800.00 | **Your 7% Gross Fee:** ₹56.00 | **Organizer's 93% Share:** ₹744.00.
> **Razorpay 2% Cut:** ₹16.00 + **18% GST on Razorpay Cut:** ₹2.88 = **₹18.88 Total Gateway Cost (2.36%)**.
> **Your Pure Net Profit per Ticket:** ₹56.00 − ₹18.88 = **₹37.12 (4.64% Net Profit)**.
> *Bonus Feature — "Pass Fee to Buyer" Toggle:* If an organizer wants to keep 100% (all ₹800), they flip one switch so the buyer pays ₹856 at checkout (₹800 ticket + ₹56 convenience fee). You still earn ₹37.12 net profit!

Local Event Type & Scale	Ticket Price	Tickets Sold	Total Sales (GMV*)	Total 7% Fee	Razorpay (2.36%)	Your Net Profit (4.64%)
Coaching Seminar / Test	₹100	500 Students	₹50,000	₹3,500	₹1,180	**₹2,320**
College Fest / Cultural Show	₹300	800 Attendees	₹2,40,000	₹16,800	₹5,664	**₹11,136**
Medium New Year Party	₹800	500 Attendees	₹4,00,000	₹28,000	₹9,440	**₹18,560**
Large Dandiya / Star Night	₹800	1,000 Attendees	₹8,00,000	₹56,000	₹18,880	**₹37,120**
Combined Season Total (4 Events)	Mixed	2,800 Tickets	₹14,90,000	₹1,04,300	₹35,164	**₹69,136 (₹0 Server Cost)**

Note: **GMV (Gross Merchandise Value) means the total value of all tickets sold through your website before deducting fees.*

5. COMPLETE PRODUCT FEATURE SCOPE (WHAT IS IN-SCOPE)

- **60-Second Event Creator:** A clean single-page form where an organizer enters the Event Title, Venue in Sasaram, Date/Time, Poster Image, Ticket Price, Total Seats, and their WhatsApp number.
- **WhatsApp-Friendly Preview Cards:** Uses **OpenGraph Tags** (special hidden webpage labels that make a link show a rich poster image, title, and price automatically when shared inside WhatsApp groups).
- **2-Step No-Login UPI Checkout:** Buyers never create an account or remember passwords. They simply enter their Name + WhatsApp Number, pick ticket quantity, and pay via Google Pay, PhonePe, or Paytm.
- **Custom Coaching & College Questions:** Organizers can ask custom fields during checkout—such as *"Enter Class / Batch / Roll Number"* for coaching scholarship tests or college events.
- **Tamper-Proof QR Ticket Card:** Immediately after payment, the screen displays a downloadable pass with a **Cryptographic QR Code** (a barcode locked with a secret mathematical key so no student can photoshop or fake a pass).
- **4-Digit PIN Mobile Gate Scanner:** Gate guards open a special webpage on their phone camera, enter a 4-digit PIN, and scan QR codes in under 1 second. If a screenshot is scanned twice, the screen flashes bright RED ("ALREADY USED").
- **Offline Cash-at-Counter Promoter Button & Live Audience Polls:** Reception desks can accept physical cash and generate an instant QR pass on the spot, plus run live audience voting during the seminar or fest.

6. HOW WE APPROACH THE BUILD: REVERSE-ENGINEERING STRATEGY

Beginners try to write every single screen from zero and quit after six months. Smart founders use **Reverse Engineering** (studying existing, battle-tested apps and open-source software by taking them apart to understand how their screens, payment flows, and security work, and then rebuilding the best parts into our own unified platform).

The Golden Rule of Our Build: Different open-source projects are written in different computer languages (like PHP, Python, or Node.js). You must **never stitch different languages together** like a messy patchwork. Instead, we use **Next.js + TypeScript + Supabase** as our single foundation, and we reverse-engineer specific features from existing tools.

> **REVERSE-ENGINEERING TOOLS & LEARNING REPOSITORIES WE USE**
>
> **1. Ghidra** (`https://ghidra-sre.org/`): A famous free **Software Reverse Engineering (SRE) Suite & Decompiler** (a tool created by the US National Security Agency that takes compiled software or mobile app files and translates them back into readable logic). Useful for understanding how compiled mobile check-in apps verify offline tickets.
>
> **2. Hopper Disassembler** (`https://www.hopperapp.com/`): A reverse-engineering **Disassembler** (a program that lets you inspect the internal machine instructions and secret workflows inside compiled Mac/Linux/iOS apps).
>
> **3. Reverse Engineer Anything Guides** (`https://github.com/mytechnotalent/Reverse-Engineering` & **Chrome DevTools Network Inspector**): While Ghidra and Hopper are used for compiled binary apps, 95% of our speed comes from inspecting live web apps using your browser's *"Inspect Element & Network Tab"* and open-source **GitHub Repositories** (public folders on GitHub.com where developers share 100% of their original source code for free).

7. THE 6 OPEN-SOURCE REPOSITORIES & WEB APPS WE REVERSE-ENGINEER

Project Name & Official Links	Original Tech	Exact Workflows & Features We Reverse-Engineer Into SSM Events
1. Sale-Ticket (TicketFlow) `https://github.com/abdelhamid1111111/sale-ticket`	Next.js, TypeScript, Supabase, Tailwind	**Our Base Starter Template:** We clone this repository first! Copy its 2-step no-login checkout screen (Buyer Name/Phone + Quantity), session-based "My Tickets" screen, and the Admin Revenue & Orders dashboard.
2. Tickity `https://github.com/akash202004/tickity`	Next.js 14, Tailwind, QR Tickets	**QR Ticket Card & Queue Logic:** Study and copy its visual mobile QR pass layout (how the ticket looks when downloaded on a phone) and its waiting-room queue system for high-demand ticket drops.
3. Hi.Events Web: `https://hi.events/` Repo: `https://github.com/HiEventsDev/Hi.Events`	PHP (Laravel), React, PostgreSQL	**Gate Check-In Scanner & Promo Codes:** Inspect their *checkin-web* module to see how a mobile phone browser camera scans QR tickets at the venue gate and how Early-Bird / VIP ticket tiers are organized.
4. Revel Web: `https://letsrevel.io/` Repo: `https://github.com/letsrevel/revel-frontend` `https://github.com/letsrevel/revel-backend`	Python (Django) + SvelteKit	**Coaching & College Custom Forms:** Reverse-engineer how Revel builds custom RSVP questionnaires so Sasaram coaching institutes can ask students for their *Class, Batch, Father's Name, or Roll No.* during checkout.
5. Gathio Web: `https://gath.io/new` Repo: `https://github.com/lowercasename/gathio`	Node.js, Express, MongoDB	**60-Second Event Creator & WhatsApp Previews:** Study Gathio's ultra-simple single-page event creation form and how it generates clean preview cards when an event link is pasted into WhatsApp.
6. Pollaris Web: `https://pollaris.org`	Open-Source Web Polling Tool	**Live Audience Q&A / Voting Tab:** Reverse-engineer its simple live voting design so coaching teachers can run live quiz polls and

8. OUR TOTAL ₹0/MONTH TECH STACK & TOOL LINKS (EXPLAINED SIMPLY)

In business, keeping your **Fixed Overhead Costs** (monthly expenses you must pay even if you sell zero tickets, like rent or server bills) at **₹0** means you can never go bankrupt. Every tool below has a generous **100% Free Tier**:

- **1. Core Website Framework — Next.js & TypeScript** (`https://nextjs.org` & `https://www.typescriptlang.org`): Next.js is the modern web framework that builds both what users see **(Frontend)** and the secret payment brain behind the scenes **(Backend)**. TypeScript checks for spelling or math mistakes before your site goes live.
- **2. Free Website Hosting — Vercel Hobby Tier** (`https://vercel.com`): The cloud company that puts your website on the internet for ₹0/month. It uses **Serverless Functions** (mini cloud computers that wake up instantly in 0.01 seconds whenever someone buys a ticket).
- **3. Main Database — Supabase Free Tier** (`https://supabase.com`): A free cloud **PostgreSQL Database** (an ultra-secure digital ledger that functions like an indestructible Excel sheet in the cloud). Its free 500MB limit can store over **5,000 events and 5,00,000+ tickets**!
- **4. Speed Lock & Gate Scanner Memory — Upstash Redis** (`https://upstash.com`): Handles **Concurrency** (what happens when 50 students click "Buy" on the last 5 seats at the exact same second—Redis locks the seats so you never oversell) and verifies QR scans at the gate in under 0.05 seconds.
- **5. Poster Image Compression — Cloudinary Free Tier** (`https://cloudinary.com`): Automatically shrinks heavy 5MB event posters into tiny 50KB **WebP Images** so your website loads fast even on weak Jio/Airtel 4G signal inside crowded marriage halls in Sasaram.
- **6. Security Shield & Wake-Up Timer — Cloudflare Free Tier** (`https://www.cloudflare.com`): Provides a **CDN (Content Delivery Network)** and **DDoS Protection** (stops hackers or fake traffic spikes from crashing your site) plus free automated timers.
- **7. UPI Payment Gateways — Razorpay & Cashfree** (`https://razorpay.com` & `https://www.cashfree.com`): Razorpay is our primary gateway for UPI payments. Cashfree is our backup gateway for automated bank splits later. Both cost ₹0/month to set up—they only take their 2% fee when you make a real sale!

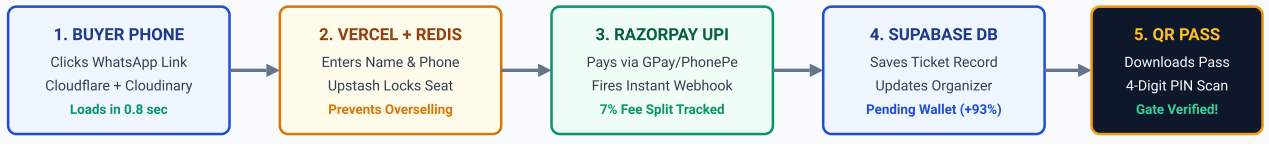


Figure 1: Complete End-to-End Ticket Purchase & Verification Architecture (₹0/Month Cloud Stack)

9. TWO DEADLY "FREE-TIER TRAPS" & HOW WE DEFEAT THEM

> **TRAP 1: WHY WE NEVER USE RENDER.COM FREE TIER (THE WEBHOOK TIMEOUT TRAP)**
>
> Many YouTube tutorials tell beginners to host backends on Render's free tier. **Do not do this!** Render puts free servers to "sleep" after 15 minutes of quiet time. When a student in Sasaram pays via UPI, Razorpay sends a **Webhook** (an automatic instant message from Razorpay's computer to your website saying *"Payment received! Issue the ticket now!"*). If your server takes 50 seconds to wake up, Razorpay gives up, the buyer's money is deducted, and **no ticket is shown!** By using **Vercel** instead, our payment receiver wakes up in 0.01 seconds every time.

> **TRAP 2: STOPPING SUPABASE FREE TIER FROM PAUSING AFTER 7 QUIET DAYS**
>
> Supabase pauses free databases if nobody visits your website for 7 straight days (for example, during exam week in Sasaram). **Our Fix:** We set up a free **Cloudflare Cron Trigger** (an automatic digital alarm clock) that quietly pings our database once every 3 days so it stays awake 24/7/365 forever!

10. COMPLETE LEGAL, BANKING & TAX GUIDE FOR A 17-YEAR-OLD FOUNDER

As a Class 11 Commerce student, you learn a foundational law in Business Studies: under **Section 11 of the Indian Contract Act, 1872**, a citizen under 18 years of age is a **Minor** and cannot legally sign a commercial contract on their own. Payment gateways like Razorpay and Cashfree follow strict Reserve Bank of India (RBI) **KYC (Know Your Customer)** rules—if you upload a 17-year-old's PAN card, their automated computer will reject your merchant account immediately.

How Do Teenage Founders in India Solve This Legally? The "Parent Guardian Proprietorship Model": Until the day you turn 18, you register the business as a **Sole Proprietorship** (a simple 1-person business where the business and the owner are legally the same person) in the name of **one trusted parent (either your Mother or your Father)**. Your parent is the official Legal Proprietor on paper (Aadhaar, PAN, and Bank Account), while you run 100% of the software, sales, and operations. The day you turn 18 and get your adult PAN card, you can smoothly transition it to your own name!

> **CRITICAL WARNING: NEVER ROTATE DIFFERENT FAMILY MEMBERS' PANS OR BANK ACCOUNTS!**
> Some people think: *"To avoid taxes, I will use my father's bank account for ₹5 Lakhs, then switch my website to my mother's PAN for ₹5 Lakhs, then my uncle's!"* **NEVER DO THIS!**
> 1. **Income Tax AIS Tracking:** Every bank account in India reports directly to a person's PAN card **AIS (Annual Information Statement)**—the tax department's master computer dashboard. Splitting money across 4 bank accounts under one PAN still adds up together automatically.
> 2. **Razorpay 180-Day Fraud Freeze:** If you suddenly swap Razorpay keys to a different family member's PAN on the same website domain, Razorpay's security system flags your site for *"Aggregator Laundering"* and freezes all your money for 180 days! Always stick to **one single parent's PAN** and follow our 2-Stage Legal Roadmap below.

Legal Stage	Ticket Sales Volume (GMV)	Exact Legal Setup & Why It Protects You
Stage 1: Direct Collection Mode	**₹0 to ₹18 Lakhs** in total ticket sales	In Bihar, **GST Registration** is only mandatory once annual business turnover crosses ₹20 Lakhs. You operate with **₹0 GST registration cost** using a free Udyam Certificate + GST Non-Enrollment Declaration + 1-Page Collection Agent Agreement.
Stage 2: Automated Gateway Split	**Above ₹18 Lakhs** in total ticket sales	By the time you process ₹18 Lakhs in tickets, you will have earned **~₹83,500+ in pure net profit**! You then spend ₹1,500 from your profits to hire a local CA in Sasaram for GST registration and turn on **Razorpay Route / Cashfree Easy Split**.
Why Stage 2 Is Magic	Unlimited Scale Across Bihar	Once Automated Split is on, when a buyer pays ₹800, Razorpay automatically sends ₹744 (93%) straight to the Organizer's bank/PAN and only ₹56 (7%) to your bank/PAN—so only your 7% commission ever counts as your turnover!

11. STEP-BY-STEP: HOW TO REGISTER AT ₹0 COST IN 15 MINUTES (WITH YOUR PARENT)

1. **Open a Separate Savings or Current Bank Account (Your Parent's Name):** Never mix household grocery UPI payments with ticket sales! Use one clean bank account belonging to your parent so your Class 11 Accountancy ledger stays 100% neat.
2. **Get a Free Government Udyam MSME Certificate (₹0 Cost, 10 Minutes):** Sit with your parent and open the official Government of India portal: `https://udyamregistration.gov.in` *(Warning: never use fake private websites that charge ₹500—the official `.gov.in` site is 100% free!)*. Enter your parent's Aadhaar number + OTP, PAN number, and write the business name as **"SSM Events & Ticketing"**. Download the official government PDF certificate immediately.
3. **Activate Razorpay Live KYC Without GST:** Create your account on `https://razorpay.com` under your parent's PAN, upload the Udyam MSME Certificate as your government business proof, and click the **"GST Non-Enrollment Declaration"** checkbox (declaring your turnover is under ₹20 Lakhs).
4. **Use a 1-Page "Collection Agent Agreement" With Organizers:** In Commerce, there is a big difference between a *Principal Seller* and a **Collection Agent** (someone who only collects money on behalf of an organizer and keeps a 7% commission). Before every event, get the organizer to sign a simple 1-page printed sheet stating: *"SSM Events*

12. HOW THE ORGANIZER WALLET & T+2 PAYOUT ENGINE WORKS

Organizers in Sasaram do not know what API keys or payment gateways are—and they should never have to! You handle all payments centrally, and each organizer gets a simple **Organizer Dashboard** link showing their live ticket sales and wallet balance:

- **Step 1 — Buyer Pays (Day 0):** A student buys an ₹800 ticket via UPI. The student gets their QR ticket immediately. Inside the Organizer's dashboard, **Pending Settlement Balance** increases by **+₹744.00 (93%)**.
- **Step 2 — Bank Settlement (T+2 Days — Transaction Day plus 2 Business Days):** Razorpay deposits ₹781.12 into your bank account (₹800 minus ₹18.88 Razorpay fee + GST). The organizer's ₹744.00 automatically moves from *Pending* to **Available to Withdraw**.
- **Step 3 — Organizer Clicks "Request Withdrawal" & You Pay Out:** The organizer types their Bank Account + IFSC (or UPI ID) and clicks Withdraw. For your first 15 events in Sasaram, you transfer their ₹744 via NEFT/IMPS/UPI from your bank app and paste the 12-digit **UTR Number** (Unique Transaction Reference—the permanent banking proof number generated for every bank transfer in India) into the dashboard!

> **THREE GOLDEN FINANCIAL SAFETY RULES (NEVER BREAK THESE!)**
> **Rule 1 (Never Pay Unsettled Money):** Only allow withdrawals on money that has already reached your bank account after Razorpay's T+2 settlement.
> **Rule 2 (The 15% Event Holdback Rule):** Let verified local organizers withdraw 85% before the event to pay tent, DJ, or hall vendors, and hold back 15% until the morning after the event finishes so no scam organizer can cancel an event and run away with buyers' money.
> **Rule 3 (Auto-Generate UTR Settlement PDF):** Every payout prints a 1-page PDF receipt showing Gross Sales, Less 7% Commission, Net Transferred, and the 12-digit Bank UTR Number for your tax folder.

13. YOUR 14-DAY STEP-BY-STEP EXECUTION & LAUNCH PLAN IN SASARAM

Timeline	Phase Focus	Exact Action Steps You Will Take
Days 1–3	**Base Setup & Repo Clone**	Create free accounts on GitHub, Vercel (`vercel.com`), and Supabase (`supabase.com`). Clone our base starter repository (`github.com/abdelhamid1111111/sale-ticket`) and connect it to your Supabase database.
Days 4–6	**Razorpay Test Mode & 7% Math**	Sign up on `razorpay.com` and switch to **Test Mode** (lets you test fake UPI payments without real money). Add the 7% commission calculator and test the 2-step no-login checkout inspired by Gathio (`gath.io/new`).
Days 7–9	**QR Ticket Card & Gate Scanner**	Add the downloadable QR Ticket screen (inspired by Tickity) and build the 4-digit PIN mobile camera gate scanner page (inspired by `hi.events`) plus custom student Roll No./Batch fields (inspired by `letsrevel.io`).
Days 10–11	**Organizer Wallet & Cash Button**	Add the Organizer Dashboard showing Tickets Sold, Pending vs. Available Balance, the "Request Withdrawal" button, the Offline Cash-at-Counter pass button, and the live audience poll tab (`pollaris.org`).
Days 12–14	**Parent KYC & First Sasaram Pitch**	Sit with your parent to get the free Udyam Certificate (`udyamregistration.gov.in`), complete Razorpay Live KYC, and walk into 3 top coaching institutes or event organizers in Sasaram (Gaurakshani / Post Office Chowk area) with a live demo on your phone!

> **HOW TO PITCH YOUR FIRST COACHING DIRECTOR IN SASARAM (WORD-FOR-WORD SCRIPT)**
> *"Sir, for your upcoming scholarship test or seminar, students won't have to stand in long queues to buy paper forms. Put this single link on your WhatsApp status—students enter their Name, Class, and Roll Number, pay via UPI in 10 seconds without any login, and get a tamper-proof QR admit card on their phone. At the gate, your guard scans their pass in 1 second using a 4-digit PIN. If you pass the 7% convenience fee to the buyer, it costs your institute ₹0!"*